# Hwang 2022 Pancreatic Ductal Adenocarcinoma — analysis summary

RUN 3 OF 4

## GO / NO-GO GATES

### Gate 1 — mixing & continuity → GO

29/29 samples (≥100 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 98 cells / 6 specks detached; iLISI median 1.00→9.31 of 29 samples); UMAP axis ratio 1.37 (compact mass, not a strip of islands). Proliferation is not a confound (module·cycling |ρ|≤0.05).

### Gate 2 — module geography → GO (non-branching)

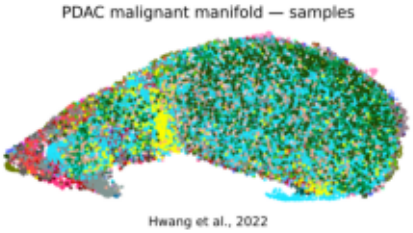
Single organized cross-repression axis, not a bifurcation: Gast↔pSom mutually exclusive (smoothed r=−0.32; joint Gast&pSom-high enrichment 0.44 — no co-expression hub). pSom high at the origin declines along pseudotime (ρ=−0.34), Gast rises (ρ=+0.29); neural crest is a terminal / inverse readout (crest-high decile mean pt=0.63), anti-correlated with both modules (vs pSom r=−0.39, vs Gast r=−0.30 — not a crest-arm inversion).

### Gate 3 — hub mutual exclusivity → NA

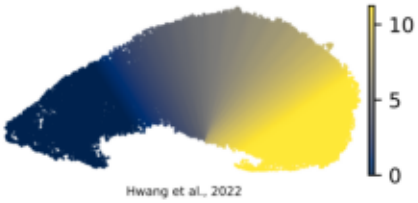
Single monotone axis — no bifurcation and no Gast/pSom co-expression hub, so there is no branch pair to test for competitive cross-repression. Branching-only correlation / modules steps are not produced.

## STEP FIGURES

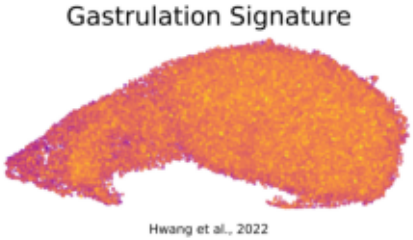
### Manifold — samples



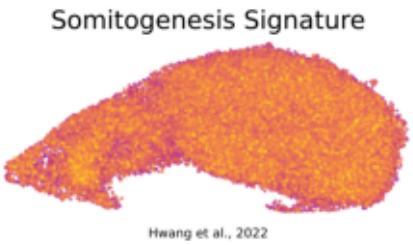

### Pseudotime



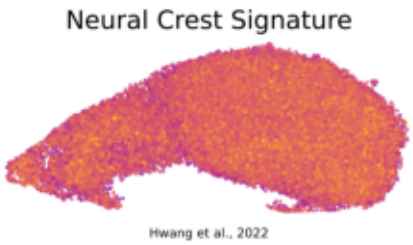

### Gastrulation



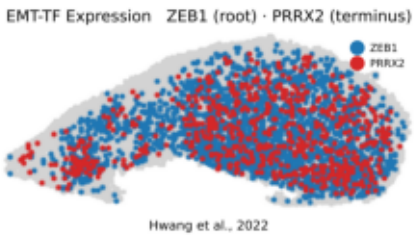

### Somitogenesis




### Neural Crest




### EMT-TF expression




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## Dataset.

PDAC (Hwang et al., 2022), snRNA-seq TP10K — 43 samples / 225k nuclei across 3 group matrices; malignant only: 64,538 nuclei → 64,104 across 29 samples ≥100 cells → 64,006 after island removal enter the trajectory.

## Verdict.

Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Hwang's PDAC malignant manifold is a single continuous axis rooted at a somitogenesis-high / neural-crest-low origin, with organized Gast↔pSom cross-repression and neural crest as a terminal inverse gradient — the developmental axis without Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The curve is fold-free (2 tips, 0 forks; self-approach 2.81) and continuous on an independent Harmony-PCA geometry (ρ=0.73; centroid-path ratio 2.27). The axis spans both transcriptional compartments (Basal-like 54,795 / Classical-like 9,211) and is orthogonal to cell cycle (93% non-cycling), so the whole manifold is kept. Raw module means are low-amplitude (Gast −0.004 / pSom −0.002 / Crest +0.003) but the geography is decisive. EMT-TFs ZEB1 (early/root) → PRRX2 (late/terminus); ZEB1 dominant (15.7%).

## Parameter choices

| | | | |
|---|---|---|---|
| **Min cells/sample** | 100 floor (29/29 kept) | **Integration** | Harmony θ=10; n_pcs=20 |
| **Scope** | whole manifold (no scoping) | **Trajectory** | scFates curve N=20, λ=0.3, μ=100 |
| **Root node** | tip 2 = pSom-high, Crest-low | **Compartments** | Basal-like 54,795 / Classical-like 9,211 |
| **Geometry** | Linear (non-branching) | **EMT-TFs** | ZEB1 (root) → PRRX2 (terminus); ZEB1 dominant 15.7% |